Luthira Peiris

Software Engineering Undergraduate — Full-Stack Developer — Cloud & DevOps Enthusiast
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LinkedIn GitHub Portfolio Medium

Summary

Software Engineering undergraduate with hands-on experience in full-stack web application development, REST API development, databases, cloud computing, and DevOps fundamentals. Skilled in building practical software solutions using React.js, TypeScript, Node.js, Express.js, MySQL, Firebase, Git, Docker, Jenkins, AWS, and Trivy. Interested in contributing to enterprise web applications, backend APIs, frontend components, testing, debugging, CI/CD workflows, and AI-assisted software development as a Software Engineer Intern.

Education

BSc (Hons) in Software Engineering July 2024 – Expected 2028
Sabaragamuwa University of Sri Lanka

Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Database Management Systems, Web Development, Software Engineering, Cloud Computing, DevOps Fundamentals

Technical Skills

Languages: JavaScript, TypeScript, Python, Java
Frontend: React.js, HTML, CSS, Tailwind CSS, Responsive UI Development
Backend / APIs: Node.js, Express.js, REST APIs, JWT Authentication, API Integration
Databases: MySQL, SQL, Firebase Firestore, DBMS Concepts, Relational Database Design
DevOps & Cloud: Jenkins, Docker, AWS EC2, Amazon VPC, GitHub Webhooks, CI/CD Fundamentals, Trivy
Tools: Git, GitHub, Visual Studio Code, Postman, Arduino IDE
AI-Assisted Development: ChatGPT, AI-assisted debugging, code explanation, documentation support

Projects

CollabSolve – Research Collaboration Platform *React.js, Node.js, Express.js, MySQL, JWT, REST APIs*

- Developed a full-stack research collaboration platform for students, researchers, and engineers to post technical problems, propose solutions, collaborate through discussions, and preserve verified solutions as reusable knowledge.
- Designed and implemented REST APIs for authentication, problem posts, comments, solutions, saved posts, notifications, user profiles, reputation, leaderboards, badges, and administrative management.
- Implemented an AI-assisted solution verification feature to evaluate submitted solutions and support the validation of answers before solved problems are archived.
- Built user-focused features including the User Dashboard, My Solutions tracking, saved problems, notifications, User Profiles, and a reputation-based Leaderboard.
- Implemented JWT-based authentication and role-based access control to secure user, administrator, and protected application routes.
- Designed and managed a relational MySQL database covering users, research fields, posts, comments, solutions, badges, reputation, notifications, saved posts, and the verified knowledge archive.
- Added dark, light, and system theme support and designed responsive interfaces to improve usability across the collaboration and problem-solving workflow.

LabourX – Construction Technician Marketplace *React, TypeScript, Firebase, Firestore, Tailwind CSS*

- Developed a marketplace platform that connects construction clients with technicians such as electricians, plumbers, masons, architects, and interior designers.
- Built user, technician, and admin interfaces for job posts, technician profiles, advertisements, boost requests, donations, and platform analytics.
- Implemented Firestore-based data handling for posts, users, boost requests, advertisements, donations, and platform activity.
- Created admin dashboard features to monitor active users, professionals, open jobs, active ads, pending boosts, donations, and platform health metrics.
- Developed monetization-related features including profile boosts, work post boosts, advertisements, and donations.

Secure CI/CD Pipeline for Web Application Deployment *Jenkins, Docker, GitHub, AWS EC2, Trivy, GitHub Webhooks*

- Built a CI/CD pipeline to automate code fetching, application build, Docker image preparation, and security scanning using Jenkins.
- Configured Jenkins on AWS EC2 and connected GitHub Webhooks to trigger automated pipeline builds after code changes.
- Used Docker to containerize application components and practiced secure deployment workflows.
- Integrated Trivy security scanning to identify vulnerabilities before pushing Docker images.
- Configured AWS networking concepts including EC2 security groups, Amazon VPC, inbound rules, and application ports.

FocusFlow – Smart IoT Focus Timer with Cloud Sync *ESP32, Firebase, Flutter, IoT, Cloud Sync*

- Built a custom IoT desktop focus timer using ESP32 to help students start, pause, and track study sessions through a physical device.
- Integrated Firebase for real-time cloud synchronization of focus session data, daily summaries, and productivity records.
- Developed a Flutter mobile app to view study sessions, track productivity progress, and interact with the device.
- Implemented hardware-based controls for managing focus sessions and syncing session updates between the device, cloud database, and mobile app.
- Designed the system to combine IoT hardware, cloud services, and mobile application development into one practical productivity solution.

Experience

CAD Designer & 3D Modeler August 2023 – January 2026
Githmi Peiris Designs, Wennappuwa, Sri Lanka

- Created 3D architectural models using SketchUp for design visualization.
- Designed 2D floor plans and technical layouts using AutoCAD.
- Converted design concepts into structured technical drawings and presentation-ready models.
- Developed attention to detail, technical documentation, and client communication skills through design-based project work.

Certifications

- Terraform for the Absolute Beginner with Hands-on Labs
- Terraform for Absolute Beginners
- Getting Started with DevOps on AWS
- AWS Academy Graduate - AWS Academy Cloud Foundations

Honors & Awards

- **5th Place** – Codemania V6.0 Competition
- **Top 15 Finalist** – CryptX 2.0 Hackathon Competition

Leadership & Volunteering

- Organizing Committee Member, Program Team – IndustriX, IEEE Student Branch of SUSL
- Organizing Committee Member, Program Team – SabraXtreme 3.0, IEEE Student Branch of SUSL
- Organizing Committee Member, Logistic Team – IEEE Day Celebration 2025, IEEE Student Branch of SUSL
- Organizing Committee Member, Content Writing Team – Beyond the Resume, Rotaract Club of SUSL